

```
In [1]: import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import MobileNetV3Small
from tensorflow.keras.models import Model
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Dropout
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau

from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns

# ignoring warnings
import warnings
warnings.simplefilter("ignore")
```

```
In [2]: # Parámetros generales
image_height = 224
image_width = 224
batch_size = 32
epochs = 20
input_shape = (image_height, image_width, 3)
num_classes = None # Se infiere del generador

data_dir = "Datos Finales Estado Balanceados"
metadata_file = "archivos_datos_balanceados_estado.csv"

df = pd.read_csv(metadata_file, sep=";")
df = df.rename(columns={'Nombre del archivo': 'image_id'})
df['Planta_Estado'] = df['Planta_Estado'].astype(str)

categoria = 'Planta_Estado'
image_id = 'image_id'
```

```
In [3]: # Dividir en entrenamiento y validación
from sklearn.model_selection import train_test_split

train_df, test_df = train_test_split(df, test_size=0.2, stratify=df['Planta_Estado'],
train_df.shape, test_df.shape)
```

```
Out[3]: ((50400, 7), (12601, 7))
```

```
In [4]: # Total de clases
print("Clases:", df['Planta_Estado'].nunique())
```

```
num_classes = df['Planta_Estado'].nunique()
class_names = df['Planta_Estado'].unique().tolist()
print(class_names)
```

Clases: 18

```
['Maíz_Enferma', 'Tomate_Saludable', 'Yuca_Enferma', 'Pepino Cohombro_Enferma', 'Tomate_Enferma', 'Manzana_Saludable', 'Fresa_Enferma', 'Maíz_Saludable', 'Papa_Enferma', 'Fresa_Saludable', 'Coliflor_Enferma', 'Pepino Cohombro_Saludable', 'Yuca_Saludable', 'Fríjol_Enferma', 'Papa_Saludable', 'Fríjol_Saludable', 'Coliflor_Saludable', 'Manzana_Enferma']
```

```
In [5]: train_df, val_df = train_test_split(train_df, test_size=0.2, stratify=train_df['Planta_Estado'],
train_df.shape, val_df.shape)
```

```
Out[5]: ((40320, 7), (10080, 7))
```

```
In [6]: # Crear el generador de imágenes --
datagen = ImageDataGenerator(
    validation_split = 0.2,
    rescale=1./255,
    preprocessing_function = None,
    rotation_range = 30,
    zoom_range = 0.2,
    horizontal_flip = True,
    vertical_flip = True,
    fill_mode = 'nearest',
    shear_range = 0.1,
    height_shift_range = 0.1,
    width_shift_range = 0.1
)

train_generator = datagen.flow_from_dataframe(
    dataframe=train_df,
    directory=data_dir,
    x_col=image_id,
    y_col=categoria,
    target_size=(image_height, image_width),
    batch_size=batch_size,
    class_mode='categorical', # Importante para clasificación multiclase
    subset='training',
    seed=42,
    classes=class_names # Asegurarse del orden de las clases
)

validation_generator = datagen.flow_from_dataframe(
    dataframe=val_df,
    directory=data_dir,
    x_col=image_id,
    y_col=categoria,
    target_size=(image_height, image_width),
```

```

    batch_size=batch_size,
    class_mode='categorical',
    subset='validation',
    seed=42,
    classes=class_names
)

```

Found 32256 validated image filenames belonging to 18 classes.

Found 2015 validated image filenames belonging to 18 classes.

```

In [7]: base_model = MobileNetV3Small(include_top=False, weights='imagenet', input_s
base_model.trainable = False

x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dropout(0.3)(x)
output = Dense(num_classes, activation='softmax')(x)

model = Model(inputs=base_model.input, outputs=output)
model.compile(optimizer=Adam(learning_rate=1e-4), loss='categorical_crossentropy',
model.summary()

```

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	–
rescaling (Rescaling)	(None, 224, 224, 3)	0	input_layer[0]
conv (Conv2D)	(None, 112, 112, 16)	432	rescaling[0][0]
conv_bn (BatchNormalization)	(None, 112, 112, 16)	64	conv[0][0]
activation (Activation)	(None, 112, 112, 16)	0	conv_bn[0][0]
expanded_conv_dept... (ZeroPadding2D)	(None, 113, 113, 16)	0	activation[0]
expanded_conv_dept... (DepthwiseConv2D)	(None, 56, 56, 16)	144	expanded_conv...
expanded_conv_dept... (BatchNormalization)	(None, 56, 56, 16)	64	expanded_conv...
re_lu (ReLU)	(None, 56, 56, 16)	0	expanded_conv...

expanded_conv_sque... (GlobalAveragePool...	(None, 1, 1, 16)	0	re_lu[0][0]
expanded_conv_sque... (Conv2D)	(None, 1, 1, 8)	136	expanded_conv
expanded_conv_sque... (ReLU)	(None, 1, 1, 8)	0	expanded_conv
expanded_conv_sque... (Conv2D)	(None, 1, 1, 16)	144	expanded_conv
add (Add)	(None, 1, 1, 16)	0	expanded_conv
re_lu_1 (ReLU)	(None, 1, 1, 16)	0	add[0][0]
multiply (Multiply)	(None, 1, 1, 16)	0	re_lu_1[0][0]
expanded_conv_sque... (Multiply)	(None, 56, 56, 16)	0	re_lu[0][0], multiply[0][0]
expanded_conv_proj... (Conv2D)	(None, 56, 56, 16)	256	expanded_conv
expanded_conv_proj... (BatchNormalizatio...	(None, 56, 56, 16)	64	expanded_conv
expanded_conv_1_ex... (Conv2D)	(None, 56, 56, 72)	1,152	expanded_conv
expanded_conv_1_ex... (BatchNormalizatio...	(None, 56, 56, 72)	288	expanded_conv
re_lu_2 (ReLU)	(None, 56, 56, 72)	0	expanded_conv
expanded_conv_1_de... (ZeroPadding2D)	(None, 57, 57, 72)	0	re_lu_2[0][0]
expanded_conv_1_de... (DepthwiseConv2D)	(None, 28, 28, 72)	648	expanded_conv
expanded_conv_1_de... (BatchNormalizatio...	(None, 28, 28, 72)	288	expanded_conv
re_lu_3 (ReLU)	(None, 28, 28, 72)	0	expanded_conv
expanded_conv_1_pr... (Conv2D)	(None, 28, 28, 24)	1,728	re_lu_3[0][0]
expanded_conv_1_pr...	(None, 28, 28,	96	expanded_conv

(BatchNormalizatio...	24)		
expanded_conv_2_ex... (Conv2D)	(None, 28, 28, 88)	2,112	expanded_conv
expanded_conv_2_ex... (BatchNormalizatio...	(None, 28, 28, 88)	352	expanded_conv
re_lu_4 (ReLU)	(None, 28, 28, 88)	0	expanded_conv
expanded_conv_2_de... (DepthwiseConv2D)	(None, 28, 28, 88)	792	re_lu_4[0][0]
expanded_conv_2_de... (BatchNormalizatio...	(None, 28, 28, 88)	352	expanded_conv
re_lu_5 (ReLU)	(None, 28, 28, 88)	0	expanded_conv
expanded_conv_2_pr... (Conv2D)	(None, 28, 28, 24)	2,112	re_lu_5[0][0]
expanded_conv_2_pr... (BatchNormalizatio...	(None, 28, 28, 24)	96	expanded_conv
expanded_conv_2_add (Add)	(None, 28, 28, 24)	0	expanded_conv expanded_conv
expanded_conv_3_ex... (Conv2D)	(None, 28, 28, 96)	2,304	expanded_conv
expanded_conv_3_ex... (BatchNormalizatio...	(None, 28, 28, 96)	384	expanded_conv
activation_1 (Activation)	(None, 28, 28, 96)	0	expanded_conv
expanded_conv_3_de... (ZeroPadding2D)	(None, 31, 31, 96)	0	activation_1[
expanded_conv_3_de... (DepthwiseConv2D)	(None, 14, 14, 96)	2,400	expanded_conv
expanded_conv_3_de... (BatchNormalizatio...	(None, 14, 14, 96)	384	expanded_conv
activation_2 (Activation)	(None, 14, 14, 96)	0	expanded_conv
expanded_conv_3_sq... (GlobalAveragePool...	(None, 1, 1, 96)	0	activation_2[

expanded_conv_3_sq... (Conv2D)	(None, 1, 1, 24)	2,328	expanded_conv
expanded_conv_3_sq... (ReLU)	(None, 1, 1, 24)	0	expanded_conv
expanded_conv_3_sq... (Conv2D)	(None, 1, 1, 96)	2,400	expanded_conv
add_1 (Add)	(None, 1, 1, 96)	0	expanded_conv
re_lu_6 (ReLU)	(None, 1, 1, 96)	0	add_1[0][0]
multiply_1 (Multiply)	(None, 1, 1, 96)	0	re_lu_6[0][0]
expanded_conv_3_sq... (Multiply)	(None, 14, 14, 96)	0	activation_2[multiply_1[0]
expanded_conv_3_pr... (Conv2D)	(None, 14, 14, 40)	3,840	expanded_conv
expanded_conv_3_pr... (BatchNormalizatio...	(None, 14, 14, 40)	160	expanded_conv
expanded_conv_4_ex... (Conv2D)	(None, 14, 14, 240)	9,600	expanded_conv
expanded_conv_4_ex... (BatchNormalizatio...	(None, 14, 14, 240)	960	expanded_conv
activation_3 (Activation)	(None, 14, 14, 240)	0	expanded_conv
expanded_conv_4_de... (DepthwiseConv2D)	(None, 14, 14, 240)	6,000	activation_3[
expanded_conv_4_de... (BatchNormalizatio...	(None, 14, 14, 240)	960	expanded_conv
activation_4 (Activation)	(None, 14, 14, 240)	0	expanded_conv
expanded_conv_4_sq... (GlobalAveragePool...	(None, 1, 1, 240)	0	activation_4[
expanded_conv_4_sq... (Conv2D)	(None, 1, 1, 64)	15,424	expanded_conv
expanded_conv_4_sq... (ReLU)	(None, 1, 1, 64)	0	expanded_conv
expanded_conv_4_sq...	(None, 1, 1, 240)	15,600	expanded_conv

(Conv2D)			
add_2 (Add)	(None, 1, 1, 240)	0	expanded_conv
re_lu_7 (ReLU)	(None, 1, 1, 240)	0	add_2[0][0]
multiply_2 (Multiply)	(None, 1, 1, 240)	0	re_lu_7[0][0]
expanded_conv_4_sq... (Multiply)	(None, 14, 14, 240)	0	activation_4[multiply_2[0]
expanded_conv_4_pr... (Conv2D)	(None, 14, 14, 40)	9,600	expanded_conv
expanded_conv_4_pr... (BatchNormalizatio...	(None, 14, 14, 40)	160	expanded_conv
expanded_conv_4_add (Add)	(None, 14, 14, 40)	0	expanded_conv expanded_conv
expanded_conv_5_ex... (Conv2D)	(None, 14, 14, 240)	9,600	expanded_conv
expanded_conv_5_ex... (BatchNormalizatio...	(None, 14, 14, 240)	960	expanded_conv
activation_5 (Activation)	(None, 14, 14, 240)	0	expanded_conv
expanded_conv_5_de... (DepthwiseConv2D)	(None, 14, 14, 240)	6,000	activation_5[
expanded_conv_5_de... (BatchNormalizatio...	(None, 14, 14, 240)	960	expanded_conv
activation_6 (Activation)	(None, 14, 14, 240)	0	expanded_conv
expanded_conv_5_sq... (GlobalAveragePool...	(None, 1, 1, 240)	0	activation_6[
expanded_conv_5_sq... (Conv2D)	(None, 1, 1, 64)	15,424	expanded_conv
expanded_conv_5_sq... (ReLU)	(None, 1, 1, 64)	0	expanded_conv
expanded_conv_5_sq... (Conv2D)	(None, 1, 1, 240)	15,600	expanded_conv
add_3 (Add)	(None, 1, 1, 240)	0	expanded_conv

re_lu_8 (ReLU)	(None, 1, 1, 240)	0	add_3[0][0]
multiply_3 (Multiply)	(None, 1, 1, 240)	0	re_lu_8[0][0]
expanded_conv_5_sq... (Multiply)	(None, 14, 14, 240)	0	activation_6[multiply_3[0]
expanded_conv_5_pr... (Conv2D)	(None, 14, 14, 40)	9,600	expanded_conv
expanded_conv_5_pr... (BatchNormalizatio...	(None, 14, 14, 40)	160	expanded_conv
expanded_conv_5_add (Add)	(None, 14, 14, 40)	0	expanded_conv expanded_conv
expanded_conv_6_ex... (Conv2D)	(None, 14, 14, 120)	4,800	expanded_conv
expanded_conv_6_ex... (BatchNormalizatio...	(None, 14, 14, 120)	480	expanded_conv
activation_7 (Activation)	(None, 14, 14, 120)	0	expanded_conv
expanded_conv_6_de... (DepthwiseConv2D)	(None, 14, 14, 120)	3,000	activation_7[
expanded_conv_6_de... (BatchNormalizatio...	(None, 14, 14, 120)	480	expanded_conv
activation_8 (Activation)	(None, 14, 14, 120)	0	expanded_conv
expanded_conv_6_sq... (GlobalAveragePool...	(None, 1, 1, 120)	0	activation_8[
expanded_conv_6_sq... (Conv2D)	(None, 1, 1, 32)	3,872	expanded_conv
expanded_conv_6_sq... (ReLU)	(None, 1, 1, 32)	0	expanded_conv
expanded_conv_6_sq... (Conv2D)	(None, 1, 1, 120)	3,960	expanded_conv
add_4 (Add)	(None, 1, 1, 120)	0	expanded_conv
re_lu_9 (ReLU)	(None, 1, 1, 120)	0	add_4[0][0]
multiply_4 (Multiply)	(None, 1, 1, 120)	0	re_lu_9[0][0]

expanded_conv_6_sq... (Multiply)	(None, 14, 14, 120)	0	activation_8[multiply_4[0]
expanded_conv_6_pr... (Conv2D)	(None, 14, 14, 48)	5,760	expanded_conv
expanded_conv_6_pr... (BatchNormalizatio...	(None, 14, 14, 48)	192	expanded_conv
expanded_conv_7_ex... (Conv2D)	(None, 14, 14, 144)	6,912	expanded_conv
expanded_conv_7_ex... (BatchNormalizatio...	(None, 14, 14, 144)	576	expanded_conv
activation_9 (Activation)	(None, 14, 14, 144)	0	expanded_conv
expanded_conv_7_de... (DepthwiseConv2D)	(None, 14, 14, 144)	3,600	activation_9[
expanded_conv_7_de... (BatchNormalizatio...	(None, 14, 14, 144)	576	expanded_conv
activation_10 (Activation)	(None, 14, 14, 144)	0	expanded_conv
expanded_conv_7_sq... (GlobalAveragePool...	(None, 1, 1, 144)	0	activation_10
expanded_conv_7_sq... (Conv2D)	(None, 1, 1, 40)	5,800	expanded_conv
expanded_conv_7_sq... (ReLU)	(None, 1, 1, 40)	0	expanded_conv
expanded_conv_7_sq... (Conv2D)	(None, 1, 1, 144)	5,904	expanded_conv
add_5 (Add)	(None, 1, 1, 144)	0	expanded_conv
re_lu_10 (ReLU)	(None, 1, 1, 144)	0	add_5[0][0]
multiply_5 (Multiply)	(None, 1, 1, 144)	0	re_lu_10[0][0]
expanded_conv_7_sq... (Multiply)	(None, 14, 14, 144)	0	activation_10 multiply_5[0]
expanded_conv_7_pr... (Conv2D)	(None, 14, 14, 48)	6,912	expanded_conv

expanded_conv_7_pr... (BatchNormalizatio...	(None, 14, 14, 48)	192	expanded_conv
expanded_conv_7_add (Add)	(None, 14, 14, 48)	0	expanded_conv expanded_conv
expanded_conv_8_ex... (Conv2D)	(None, 14, 14, 288)	13,824	expanded_conv
expanded_conv_8_ex... (BatchNormalizatio...	(None, 14, 14, 288)	1,152	expanded_conv
activation_11 (Activation)	(None, 14, 14, 288)	0	expanded_conv
expanded_conv_8_de... (ZeroPadding2D)	(None, 17, 17, 288)	0	activation_11
expanded_conv_8_de... (DepthwiseConv2D)	(None, 7, 7, 288)	7,200	expanded_conv
expanded_conv_8_de... (BatchNormalizatio...	(None, 7, 7, 288)	1,152	expanded_conv
activation_12 (Activation)	(None, 7, 7, 288)	0	expanded_conv
expanded_conv_8_sq... (GlobalAveragePool...	(None, 1, 1, 288)	0	activation_12
expanded_conv_8_sq... (Conv2D)	(None, 1, 1, 72)	20,808	expanded_conv
expanded_conv_8_sq... (ReLU)	(None, 1, 1, 72)	0	expanded_conv
expanded_conv_8_sq... (Conv2D)	(None, 1, 1, 288)	21,024	expanded_conv
add_6 (Add)	(None, 1, 1, 288)	0	expanded_conv
re_lu_11 (ReLU)	(None, 1, 1, 288)	0	add_6[0][0]
multiply_6 (Multiply)	(None, 1, 1, 288)	0	re_lu_11[0][0]
expanded_conv_8_sq... (Multiply)	(None, 7, 7, 288)	0	activation_12 multiply_6[0]
expanded_conv_8_pr... (Conv2D)	(None, 7, 7, 96)	27,648	expanded_conv
expanded_conv_8_pr...	(None, 7, 7, 96)	384	expanded_conv

(BatchNormalizatio...			
expanded_conv_9_ex... (Conv2D)	(None, 7, 7, 576)	55,296	expanded_conv
expanded_conv_9_ex... (BatchNormalizatio...	(None, 7, 7, 576)	2,304	expanded_conv
activation_13 (Activation)	(None, 7, 7, 576)	0	expanded_conv
expanded_conv_9_de... (DepthwiseConv2D)	(None, 7, 7, 576)	14,400	activation_13
expanded_conv_9_de... (BatchNormalizatio...	(None, 7, 7, 576)	2,304	expanded_conv
activation_14 (Activation)	(None, 7, 7, 576)	0	expanded_conv
expanded_conv_9_sq... (GlobalAveragePool...	(None, 1, 1, 576)	0	activation_14
expanded_conv_9_sq... (Conv2D)	(None, 1, 1, 144)	83,088	expanded_conv
expanded_conv_9_sq... (ReLU)	(None, 1, 1, 144)	0	expanded_conv
expanded_conv_9_sq... (Conv2D)	(None, 1, 1, 576)	83,520	expanded_conv
add_7 (Add)	(None, 1, 1, 576)	0	expanded_conv
re_lu_12 (ReLU)	(None, 1, 1, 576)	0	add_7[0][0]
multiply_7 (Multiply)	(None, 1, 1, 576)	0	re_lu_12[0][0]
expanded_conv_9_sq... (Multiply)	(None, 7, 7, 576)	0	activation_14 multiply_7[0]
expanded_conv_9_pr... (Conv2D)	(None, 7, 7, 96)	55,296	expanded_conv
expanded_conv_9_pr... (BatchNormalizatio...	(None, 7, 7, 96)	384	expanded_conv
expanded_conv_9_add (Add)	(None, 7, 7, 96)	0	expanded_conv expanded_conv
expanded_conv_10_e... (Conv2D)	(None, 7, 7, 576)	55,296	expanded_conv

expanded_conv_10_e... (BatchNormalizatio...	(None, 7, 7, 576)	2,304	expanded_conv
activation_15 (Activation)	(None, 7, 7, 576)	0	expanded_conv
expanded_conv_10_d... (DepthwiseConv2D)	(None, 7, 7, 576)	14,400	activation_15
expanded_conv_10_d... (BatchNormalizatio...	(None, 7, 7, 576)	2,304	expanded_conv
activation_16 (Activation)	(None, 7, 7, 576)	0	expanded_conv
expanded_conv_10_s... (GlobalAveragePool...	(None, 1, 1, 576)	0	activation_16
expanded_conv_10_s... (Conv2D)	(None, 1, 1, 144)	83,088	expanded_conv
expanded_conv_10_s... (ReLU)	(None, 1, 1, 144)	0	expanded_conv
expanded_conv_10_s... (Conv2D)	(None, 1, 1, 576)	83,520	expanded_conv
add_8 (Add)	(None, 1, 1, 576)	0	expanded_conv
re_lu_13 (ReLU)	(None, 1, 1, 576)	0	add_8[0][0]
multiply_8 (Multiply)	(None, 1, 1, 576)	0	re_lu_13[0][0]
expanded_conv_10_s... (Multiply)	(None, 7, 7, 576)	0	activation_16 multiply_8[0]
expanded_conv_10_p... (Conv2D)	(None, 7, 7, 96)	55,296	expanded_conv
expanded_conv_10_p... (BatchNormalizatio...	(None, 7, 7, 96)	384	expanded_conv
expanded_conv_10_a... (Add)	(None, 7, 7, 96)	0	expanded_conv expanded_conv
conv_1 (Conv2D)	(None, 7, 7, 576)	55,296	expanded_conv
conv_1_bn (BatchNormalizatio...	(None, 7, 7, 576)	2,304	conv_1[0][0]
activation_17	(None, 7, 7, 576)	0	conv_1_bn[0][

(Activation)			
global_average_pooling2d (GlobalAveragePool2D)	(None, 576)	0	activation_17
dropout (Dropout)	(None, 576)	0	global_average
dense (Dense)	(None, 18)	10,386	dropout[0][0]

Total params: 949,506 (3.62 MB)

Trainable params: 10,386 (40.57 KB)

Non-trainable params: 939,120 (3.58 MB)

```
In [8]: early_stop = EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True)
        reduce_lr = ReduceLROnPlateau(monitor='val_loss', factor=0.2, patience=3, verbose=1)
```

```
In [9]: history = model.fit(
        train_generator,
        validation_data=validation_generator,
        epochs=epochs,
        callbacks=[early_stop, reduce_lr]
    )
```

Epoch 1/20

1008/1008 ————— **268s** 265ms/step - accuracy: 0.0610 - loss: 2.9491 - val_accuracy: 0.0883 - val_loss: 2.8741 - learning_rate: 1.0000e-04

Epoch 2/20

1008/1008 ————— **514s** 510ms/step - accuracy: 0.0659 - loss: 2.8998 - val_accuracy: 0.1261 - val_loss: 2.8550 - learning_rate: 1.0000e-04

Epoch 3/20

1008/1008 ————— **278s** 276ms/step - accuracy: 0.0790 - loss: 2.8814 - val_accuracy: 0.1285 - val_loss: 2.8424 - learning_rate: 1.0000e-04

Epoch 4/20

1008/1008 ————— **277s** 275ms/step - accuracy: 0.0904 - loss: 2.8635 - val_accuracy: 0.1906 - val_loss: 2.8272 - learning_rate: 1.0000e-04

Epoch 5/20

1008/1008 ————— **278s** 276ms/step - accuracy: 0.0962 - loss: 2.8474 - val_accuracy: 0.1464 - val_loss: 2.8122 - learning_rate: 1.0000e-04

Epoch 6/20

1008/1008 ————— **275s** 272ms/step - accuracy: 0.1001 - loss: 2.8314 - val_accuracy: 0.1464 - val_loss: 2.8026 - learning_rate: 1.0000e-04

Epoch 7/20

1008/1008 ————— **268s** 266ms/step - accuracy: 0.1072 - loss: 2.8158 - val_accuracy: 0.2015 - val_loss: 2.7900 - learning_rate: 1.0000e-04

Epoch 8/20

1008/1008 ————— **261s** 259ms/step - accuracy: 0.1126 - loss: 2.8028 - val_accuracy: 0.1687 - val_loss: 2.7780 - learning_rate: 1.0000e-04

Epoch 9/20

1008/1008 ————— **261s** 259ms/step - accuracy: 0.1153 - loss: 2.8028 - val_accuracy: 0.1687 - val_loss: 2.7780 - learning_rate: 1.0000e-04

7917 - val_accuracy: 0.1628 - val_loss: 2.7677 - learning_rate: 1.0000e-04
 Epoch 10/20
1008/1008 ————— **258s** 256ms/step - accuracy: 0.1178 - loss: 2.
 7807 - val_accuracy: 0.2010 - val_loss: 2.7580 - learning_rate: 1.0000e-04
 Epoch 11/20
1008/1008 ————— **256s** 254ms/step - accuracy: 0.1264 - loss: 2.
 7676 - val_accuracy: 0.1568 - val_loss: 2.7494 - learning_rate: 1.0000e-04
 Epoch 12/20
1008/1008 ————— **257s** 255ms/step - accuracy: 0.1211 - loss: 2.
 7616 - val_accuracy: 0.2035 - val_loss: 2.7393 - learning_rate: 1.0000e-04
 Epoch 13/20
1008/1008 ————— **259s** 257ms/step - accuracy: 0.1296 - loss: 2.
 7547 - val_accuracy: 0.2000 - val_loss: 2.7300 - learning_rate: 1.0000e-04
 Epoch 14/20
1008/1008 ————— **258s** 256ms/step - accuracy: 0.1337 - loss: 2.
 7425 - val_accuracy: 0.1881 - val_loss: 2.7255 - learning_rate: 1.0000e-04
 Epoch 15/20
1008/1008 ————— **258s** 256ms/step - accuracy: 0.1291 - loss: 2.
 7371 - val_accuracy: 0.2189 - val_loss: 2.7164 - learning_rate: 1.0000e-04
 Epoch 16/20
1008/1008 ————— **257s** 255ms/step - accuracy: 0.1409 - loss: 2.
 7284 - val_accuracy: 0.2060 - val_loss: 2.7107 - learning_rate: 1.0000e-04
 Epoch 17/20
1008/1008 ————— **341s** 339ms/step - accuracy: 0.1394 - loss: 2.
 7246 - val_accuracy: 0.2169 - val_loss: 2.7023 - learning_rate: 1.0000e-04
 Epoch 18/20
1008/1008 ————— **474s** 470ms/step - accuracy: 0.1452 - loss: 2.
 7141 - val_accuracy: 0.2233 - val_loss: 2.6979 - learning_rate: 1.0000e-04
 Epoch 19/20
1008/1008 ————— **489s** 485ms/step - accuracy: 0.1476 - loss: 2.
 7030 - val_accuracy: 0.1935 - val_loss: 2.6896 - learning_rate: 1.0000e-04
 Epoch 20/20
1008/1008 ————— **486s** 482ms/step - accuracy: 0.1434 - loss: 2.
 7030 - val_accuracy: 0.1945 - val_loss: 2.6840 - learning_rate: 1.0000e-04

```
In [10]: print("\nEvaluando el modelo...")
          evaluation_result = model.evaluate(validation_generator)
          print(f"Pérdida en la validación: {evaluation_result[0]:.4f}")
          print(f"Precisión en la validación: {evaluation_result[1]:.4f}")
```

Evaluando el modelo...

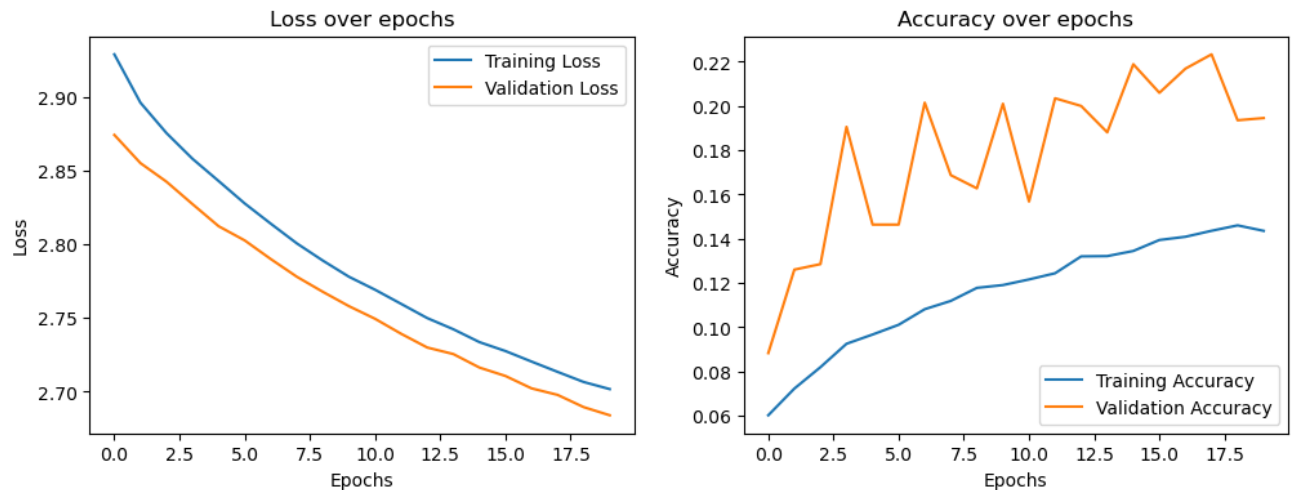
63/63 ————— **29s** 453ms/step - accuracy: 0.1907 - loss: 2.6707
 Pérdida en la validación: 2.6864
 Precisión en la validación: 0.1955

```
In [11]: # Acceder a los datos de pérdida y precisión
          loss = history.history['loss']
          val_loss = history.history['val_loss']
          accuracy = history.history['accuracy']
          val_accuracy = history.history['val_accuracy']
```

```
# Graficar la pérdida del modelo
plt.figure(figsize=(12, 4))
plt.subplot(1, 2, 1)
plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Loss over epochs')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()

# Graficar la precisión del modelo
plt.subplot(1, 2, 2)
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Accuracy over epochs')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
```

Out[11]: <matplotlib.legend.Legend at 0x28b399d60>



```
In [12]: epochs = list(range(1, 20))

loss = history.history['loss']
val_loss = history.history['val_loss']
accuracy = history.history['accuracy']
val_accuracy = history.history['val_accuracy']

data = {
    'Epoch': 20,
    'Trainning Loss': loss,
    'Validation Loss': val_loss,
    'Training Accuracy': accuracy,
    'Validation Accuracy': val_accuracy
}
```

```
}  
df = pd.DataFrame(data)  
  
# Imprimir la tabla  
print(df)
```

	Epoch	Trainning Loss	Validation Loss	Training Accuracy \
0	20	2.928769	2.874148	0.060237
1	20	2.896027	2.855033	0.072297
2	20	2.875367	2.842403	0.081969
3	20	2.857908	2.827177	0.092541
4	20	2.842866	2.812219	0.096726
5	20	2.827661	2.802592	0.101128
6	20	2.813936	2.789952	0.108166
7	20	2.800572	2.777951	0.111917
8	20	2.788871	2.767706	0.117808
9	20	2.777936	2.757995	0.119079
10	20	2.769153	2.749440	0.121621
11	20	2.759558	2.739280	0.124411
12	20	2.749935	2.729978	0.132037
13	20	2.742373	2.725543	0.132161
14	20	2.733648	2.716404	0.134487
15	20	2.727583	2.710674	0.139447
16	20	2.720447	2.702278	0.140904
17	20	2.713457	2.697854	0.143601
18	20	2.706594	2.689552	0.146050
19	20	2.701769	2.684029	0.143570

	Validation Accuracy
0	0.088337
1	0.126055
2	0.128536
3	0.190571
4	0.146402
5	0.146402
6	0.201489
7	0.168734
8	0.162779
9	0.200993
10	0.156824
11	0.203474
12	0.200000
13	0.188089
14	0.218859
15	0.205955
16	0.216873
17	0.223325
18	0.193548
19	0.194541


```
In [13]: # Accuracy final en entrenamiento
val_loss = history.history['val_loss'][-1]
```

```
In [14]: print('Perdida final en Train:', history.history['loss'][-1])
print("Pérdida final en Validation:", history.history['val_loss'][-1])

print("Accuracy final en Train:", history.history['accuracy'][-1])
print("Accuracy final en Validation:", history.history['val_accuracy'][-1])
```

Perdida final en Train: 2.701768636703491
 Pérdida final en Validation: 2.6840293407440186
 Accuracy final en Train: 0.1435701847076416
 Accuracy final en Validation: 0.19454094767570496

```
In [15]: test_df.Planta_Estado = test_df.Planta_Estado.astype('str')

test_generator = datagen.flow_from_dataframe(test_df,
      directory = data_dir,
      x_col = image_id,
      y_col = categoria,
      target_size = (image_height, image_width),
      batch_size = batch_size,
      class_mode = "categorical")
```

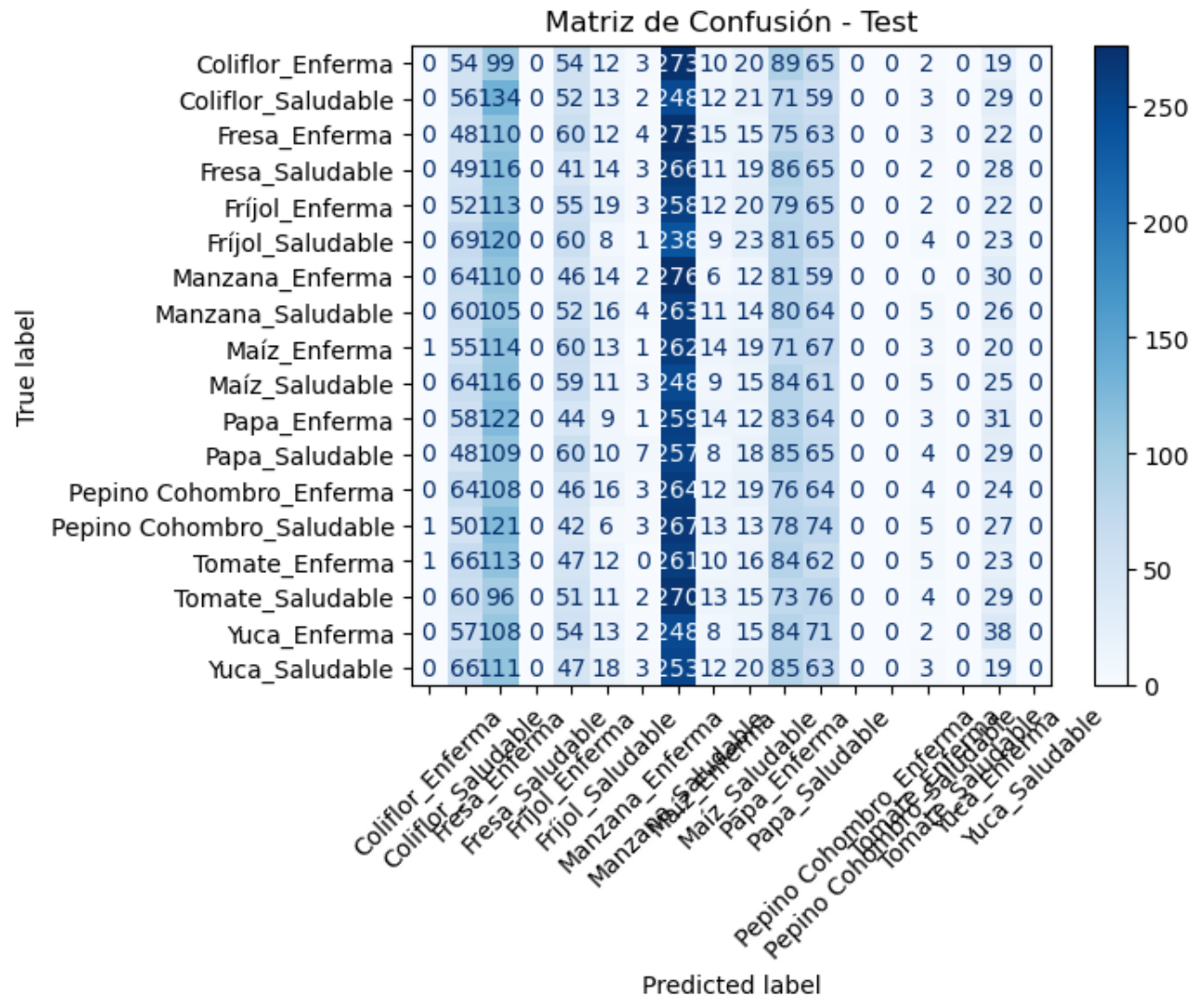
Found 12601 validated image filenames belonging to 18 classes.

```
In [16]: # -----
# 1. Obtener predicciones completas para test
# -----
y_pred_probs = model.predict(test_generator)
y_pred = np.argmax(y_pred_probs, axis=1)
y_true = test_generator.classes
```

394/394  178s 450ms/step

```
In [17]: # -----
# 2. Matriz de Confusión
# -----
from sklearn.metrics import confusion_matrix, accuracy_score, ConfusionMatrixDisplay

cm = confusion_matrix(y_true, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=test_generator.classes)
disp.plot(cmap="Blues", xticks_rotation=45)
plt.title("Matriz de Confusión - Test")
plt.show()
```



```
In [18]: from sklearn.metrics import confusion_matrix, accuracy_score, ConfusionMatrix
accuracy = accuracy_score(y_true, y_pred)
print(f"Test Accuracy: {accuracy:.4f}")
```

Test Accuracy: 0.0567

```
In [19]: from sklearn.metrics import roc_curve, auc
from tensorflow.keras.utils import to_categorical

# 3. Convertir etiquetas reales a one-hot
n_classes = y_pred_probs.shape[1]
y_true_onehot = to_categorical(y_true, num_classes=n_classes)

# 4. Calcular ROC y AUC por clase
fpr = dict()
tpr = dict()
roc_auc = dict()
```

```

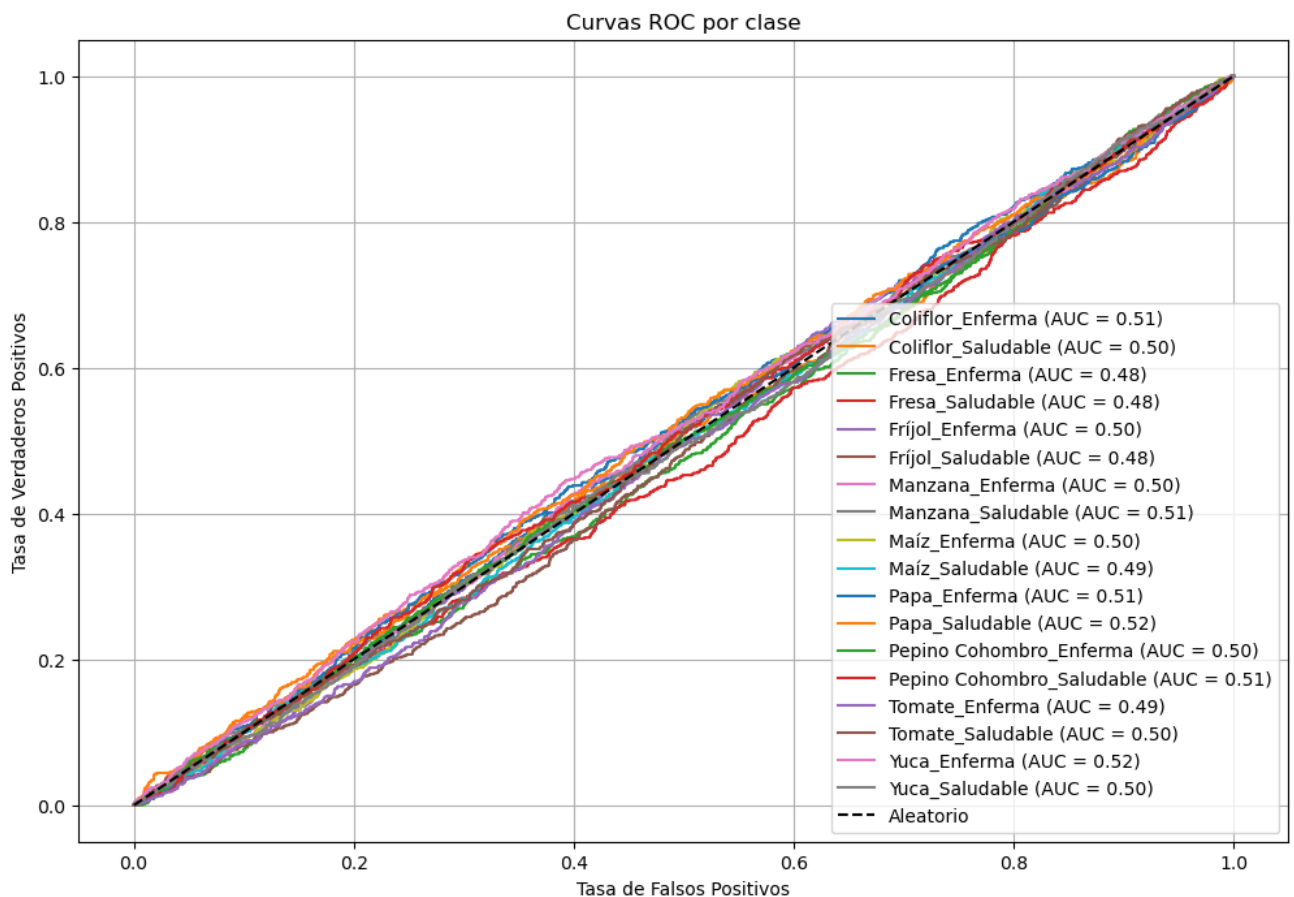
for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred_probs[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])

# 5. Obtener nombres de clase
class_names = list(test_generator.class_indices.keys())

# 6. Graficar curvas ROC
plt.figure(figsize=(12, 8))
for i in range(n_classes):
    plt.plot(fpr[i], tpr[i], label=f'{class_names[i]} (AUC = {roc_auc[i]:.2f})')

plt.plot([0, 1], [0, 1], 'k--', label='Aleatorio')
plt.xlabel('Tasa de Falsos Positivos')
plt.ylabel('Tasa de Verdaderos Positivos')
plt.title('Curvas ROC por clase')
plt.legend(loc='lower right')
plt.grid(True)
plt.show()

```



```

In [20]: from sklearn.metrics import roc_auc_score

# AUC macro (promedio de clases, todas igual de importantes)

```

```
auc_macro = roc_auc_score(y_true_onehot, y_pred_probs, average='macro')

# AUC micro (promedia sobre todas las instancias)
auc_micro = roc_auc_score(y_true_onehot, y_pred_probs, average='micro')

print(f"AUC Macro promedio: {auc_macro:.4f}")
print(f"AUC Micro promedio: {auc_micro:.4f}")
```

AUC Macro promedio: 0.4998

AUC Micro promedio: 0.5008

```
In [21]: from sklearn.metrics import f1_score
# Asegúrate de tener esto:
# y_pred = np.argmax(y_pred_probs, axis=1)
# y_true = test_generator.classes

# Accuracy
accuracy = accuracy_score(y_true, y_pred)

# F1 Scores
f1_macro = f1_score(y_true, y_pred, average='macro')
f1_micro = f1_score(y_true, y_pred, average='micro')
f1_weighted = f1_score(y_true, y_pred, average='weighted')

print(f"Accuracy: {accuracy:.4f}")
print(f"F1 Score (macro): {f1_macro:.4f}")
print(f"F1 Score (micro): {f1_micro:.4f}")
print(f"F1 Score (weighted): {f1_weighted:.4f}")

# Detalle por clase
class_names = list(test_generator.class_indices.keys())
print("\nReporte de clasificación por clase:")
print(classification_report(y_true, y_pred, target_names=class_names))
```

Accuracy: 0.0567
 F1 Score (macro): 0.0344
 F1 Score (micro): 0.0567
 F1 Score (weighted): 0.0344

Reporte de clasificación por clase:

	precision	recall	f1-score	support
Coliflor_Enferma	0.00	0.00	0.00	700
Coliflor_Saludable	0.05	0.08	0.06	700
Fresa_Enferma	0.05	0.16	0.08	700
Fresa_Saludable	0.00	0.00	0.00	700
Fríjol_Enferma	0.06	0.08	0.07	700
Fríjol_Saludable	0.04	0.01	0.02	701
Manzana_Enferma	0.04	0.00	0.01	700
Manzana_Saludable	0.06	0.38	0.10	700
Maíz_Enferma	0.07	0.02	0.03	700
Maíz_Saludable	0.05	0.02	0.03	700
Papa_Enferma	0.06	0.12	0.08	700
Papa_Saludable	0.06	0.09	0.07	700
Pepino Cohombro_Enferma	0.00	0.00	0.00	700
Pepino Cohombro_Saludable	0.00	0.00	0.00	700
Tomate_Enferma	0.08	0.01	0.01	700
Tomate_Saludable	0.00	0.00	0.00	700
Yuca_Enferma	0.08	0.05	0.07	700
Yuca_Saludable	0.00	0.00	0.00	700
accuracy			0.06	12601
macro avg	0.04	0.06	0.03	12601
weighted avg	0.04	0.06	0.03	12601

```
In [22]: model.save('modelo_18-MobileNetV3_Planta_Estadp_Balanceado.h5')
```

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save\_model(model, 'my\_model.keras')`.

```
In [23]: class_indices = train_generator.class_indices # o test_generator.class_indices
index_to_class = {v: k for k, v in class_indices.items()}
print(index_to_class)
with open("index_to_class_modelo_18-MobileNetV3_Planta_Estadp_Balanceado.json", "w") as f:
    json.dump(index_to_class, f, indent=4, ensure_ascii=False)
```

```
{0: 'Maíz_Enferma', 1: 'Tomate_Saludable', 2: 'Yuca_Enferma', 3: 'Pepino Cohombro_Enferma', 4: 'Tomate_Enferma', 5: 'Manzana_Saludable', 6: 'Fresa_Enferma', 7: 'Maíz_Saludable', 8: 'Papa_Enferma', 9: 'Fresa_Saludable', 10: 'Coliflor_Enferma', 11: 'Pepino Cohombro_Saludable', 12: 'Yuca_Saludable', 13: 'Fríjol_Enferma', 14: 'Papa_Saludable', 15: 'Fríjol_Saludable', 16: 'Coliflor_Saludable', 17: 'Manzana_Enferma'}
```

In []: